

Prathmesh Jain

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PROFESSIONAL SUMMARY

Full Stack GenAI Engineer with 1+ year of experience in Research & Innovation (CTO Office), BFSI Americas, building production-grade GenAI systems. Strong expertise in agentic AI architectures, tool-calling workflows, RAG pipelines, and hybrid LLM-deterministic systems, combined with full-stack API development.

EXPERIENCE

Tata Consultancy Services (TCS)

Oct 2024 – Present

Full Stack AI Engineer (Research & Innovation, CTO Office, BFSI Americas)

- Built and delivered end-to-end production systems for BFSI workflows, spanning **React**-based frontend applications, **FastAPI** backend services, and GenAI-driven automation.
- Designed stateful, multi-step GenAI workflows using **LangChain** and **CrewAI**, enabling controlled, iterative tool execution across search, browser automation, sanctions APIs, and internal knowledge sources.
- Integrated Playwright-based web crawling, real-time search via **Model Context Protocol (MCP)**, and **Retrieval-Augmented Generation (RAG)** to ground model outputs in external and historical data.
- Productionized AI services using FastAPI with robust reliability patterns including retries, exponential backoff, fallbacks, caching, and LLM cost controls.
- Developed frontend features in React, including Server-Sent Events (SSE), to stream and monitor long-running AI-driven workflows in real time.

PERSONAL PROJECTS

Patronix – E-commerce Platform | *React.js, Node.js, Express.js, MongoDB*

July 2023

[Live](#) | [GitHub](#)

- Built and deployed a full-stack e-commerce platform with live hosting, secure authentication, product catalog management, cart, and order tracking flows.
- Optimized application performance and global state handling using Redux Toolkit.
- Designed modular, maintainable RESTful APIs with token-based authentication and clean architecture.

ExpenseGauge – Expense Tracking App | *React Native (Expo), Node.js, MongoDB*

Dec 2025

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- Designed and shipped a cross-platform, offline-first expense tracking application with local persistence and automatic background sync on network reconnection.
- Implemented secure authentication using short-lived access tokens and rotating refresh tokens for seamless session continuity.
- Built role-based backend services supporting expense analytics with monthly trends and category-wise visualizations.

TECHNICAL SKILLS

Languages: Python, JavaScript, Java

Backend: Node.js, Express.js, FastAPI, Flask

Frontend: React.js, Next.js, React Native, Redux Toolkit, Zustand, TailwindCSS

AI / LLM: LangChain, CrewAI, MCP, Prompt Engineering, RAG, Embeddings, Tool Calling

Database: MongoDB, PostgreSQL

Tools: Docker, Git, Postman, Playwright

EDUCATION

Vidyalankar Institute of Technology

Mumbai, IN

Bachelors in Computer Engineering

May 2024

GPA: 9.2 / 10